Combinations

(Also available in WeScheme)

Students explore the concept of combinations as a restricted set of permutations in which order does not matter. They compute the possible number of outcomes when choosing n items from m possibilities in a variety of real-world situations.

Lesson Goals	Students will be able to… • Define and explain "combination" (without replacement) • Compute the number of combinations for a given situation (without replacement)
Student-facing Lesson Goals	• I can explain what combination means (without replacement) • I can compute the number of combinations from a word problem
Prerequisites	• Simple Data Types • Contracts • Functions: Contracts, Examples & Definitions • Solving Word Problems with the Design Recipe • Permutations
Materials	• Combinations (Page 28) • Combination or Permutation? (Page 29) • *Optional: Permutations and Combinations Starter File* • **Classroom visual:** Language Table

Glossary

combination :: the number of possible arrangements in a collection of items where the order of the selection does NOT matter. (ABC and CBA would be considered the same combination.)

permutation :: the number of possible arrangements in a collection of items where the order of the selection matters (ABC and CBA would be considered different permutations)

Combinations

Overview

Students are introduced to the concept of ***combinations***. They learn the definition and compute a number of combinations by hand using tree diagrams, then learn the formula for computing combinations.

Launch

With his new set of rules, Luigi's Family Restaurant is more popular than ever! But with a full house every night, the cooks are busy and the kitchen gets backed up. Customers start complaining about slow service, and Luigi is once again in trouble.

The cooks point out that it's easy to add twice as much ravioli to the same pot - but it's hard to make two *separate* pots of ravioli. If people could choose their four courses in advance, the chefs could just make a few huge batches of each item and divide them up on people's plates!

Luigi decides to change the rules again, to help his kitchen staff. There are still six items on the menu, customers can still choose any four they want, and they still can't choose the same item twice…but **order no longer matters**. Instead of a four-course meal, diners get a "combination platter" with all four items on the plate.

- How is this situation similar to the permutation questions?
- How is it different?

Investigate

When order *doesn't* matter, all the possible options are called ***combinations*** (think of Luigi's "combination platter!").

Since there's no replacement, let's start by using our formula for permutation-without-replacement to compute all the possible permutations. We have six items to choose from and are selecting four of them:

$$permute\text{-}no\text{-}replace(6,4) = \frac{6!}{(6-4)!} = \frac{6!}{2!} = 360$$

In this situation, the order *doesn't matter*, so some of these platters are going to be the same combination: Lasagna, Soup, Ziti and Chicken is a different ***permutation*** from Lasagna, Soup, Chicken, Ziti, but it's the same ***combination***! If we knew that every combination would have a duplicate, we'd divide the number of platters by two. If we knew each one would have a triplicate, we'd divide by three.

How do we compute the number of duplicates in a four-course platter?

We already know how to do this! This is the same question as "How many permutations are there for the same four courses?"

Using our formula for permutation without replacement we get: $permute\text{-}no\text{-}replace(4,4) = \frac{4!}{(4-4)!} = 4! = 24$ duplicates!

Now, we need to divide 'the number of permutations (without replacement) for choosing 4 courses out of six items" by "the number of duplicate permutations in any four-course platter".

$$combinations(6,4) = \frac{6!}{(6-4)!} \div 4! = 360 \div 24 = 15$$

We can rewrite this using our functions from earlier:

$$combinations(items, choose) = \frac{permute\text{-}no\text{-}replace(items, choose)}{permute\text{-}no\text{-}replace(choose, choose)}$$

In this situation, we have 6 possible choices and we get to choose 4 times:

$$combinations(6,4) = \frac{permute\text{-}no\text{-}replace(6,4)}{permute\text{-}no\text{-}replace(4,4)}$$

Complete the Combinations and Combination or Permutation? worksheets.

Synthesize

- What is the difference between ***combination*** and ***permutation***?
- What are some real-world examples of combinations?

Additional Exercises:

- Permutations and Combinations Starter File provides students with a chance to view all the permutations and combinations for Luigi's menu.